D1 Differential calculus II: solutions

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All final answers were checked with the computer algebra system SymPy. Try each problem seriously before reading its solution. When your answer differs only in form (for example $\frac{1}{\cos^2 x}$ against $1+\tan^2 x$), check whether the two are equal before deciding you are wrong.

Practice

1.

(a) Outer u^4, inner $3x^2-5$ with derivative $6x$: $f'(x) = 4(3x^2-5)^3 \cdot 6x = 24x(3x^2-5)^3$.

(b) $g(x) = (1+x^2)^{1/2}$, so $g'(x) = \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x = \dfrac{x}{\sqrt{1+x^2}}$.

(c) $h(x) = (2x-1)^{-3}$, so $h'(x) = -3(2x-1)^{-4} \cdot 2 = \dfrac{-6}{(2x-1)^4}$.

(d) $k'(x) = \cos(3x^2) \cdot 6x = 6x\cos(3x^2)$.

2.

(a) Product and chain rule: $f'(x) = 2xe^{-x} + x^2 \cdot (-e^{-x}) = x(2-x)e^{-x}$.

(b) $g'(x) = \dfrac{1}{x^2+1} \cdot 2x = \dfrac{2x}{x^2+1}$.

(c) $h'(x) = 2^x + x \cdot 2^x \ln 2 = 2^x(1 + x\ln 2)$.

(d) $k(x) = {}^3\log 5 + {}^3\log x$, and the first term is a constant, so $k'(x) = \dfrac{1}{x \ln 3}$.

(e) $m'(x) = \dfrac{1}{\cos^2(2x)} \cdot 2 = \dfrac{2}{\cos^2(2x)}$, or $2(1+\tan^2(2x))$.

(f) $n'(x) = e^{\sin x} \cdot \cos x$.

3.

(a) Quotient rule: $f'(x) = \dfrac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} = \dfrac{1-\ln x}{x^2}$.

(b) $f'(x) = 0 \iff \ln x = 1 \iff x = e$. Since $x^2 > 0$, the sign of f' is the sign of $1 - \ln x$: positive for $x < e$, negative for $x > e$. So there is a maximum at $(e, \frac{1}{e})$.

(c) Since $\pi > e$ and f has its maximum at e, $f(\pi) < f(e)$: $\dfrac{\ln \pi}{\pi} < \dfrac{\ln e}{e}$. Multiply by $e\pi > 0$: $e\ln\pi < \pi \ln e$, so $\ln(\pi^e) < \ln(e^\pi)$, and since ln is increasing, $\pi^e < e^\pi$. (Numerically $22.46 < 23.14$.)

4. $f'(x) = 2\sin x \cos x + 2\cos x \cdot (-\sin x) = 0$. A function whose derivative is 0 on an interval is constant there (this follows from the mean value theorem, which you will meet later; here accept it). So f is constant on $\mathbb{R}$, and its value is $f(0) = 0 + 1 = 1$. We have re-proved $\sin^2 x + \cos^2 x = 1$ using calculus alone (although of course the derivatives of sin and cos were themselves derived using circle geometry).

5.

(a) By the change-of-base rule ${}^g\log x = \dfrac{\ln x}{\ln g}$, and $\ln g$ is a nonzero constant. So $({}^g\log x)' = \dfrac{1}{\ln g}\cdot\dfrac{1}{x} = \dfrac{1}{x\ln g}$.

(b) Chain rule: $f'(x) = \dfrac{1}{(x^2-4)\ln 2}\cdot 2x$. So $f'(3) = \dfrac{6}{5\ln 2}$.

6.

(a) $2x + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x}{y}$. At $P(3,-4)$: slope $-\dfrac{3}{-4} = \dfrac{3}{4}$.

(b) The radius OP has slope $\dfrac{-4}{3}$. The product of the slopes is $\frac{3}{4}\cdot(-\frac{4}{3}) = -1$, so the lines are perpendicular. (In general the slope $-\frac{x}{y}$ is always -1 divided by the slope $\frac{y}{x}$ of the radius: the familiar fact that a tangent to a circle is perpendicular to the radius.)

7.

(a) $27 + 27 = 54 = 6\cdot 3\cdot 3$. By (b), the slope at $(3,3)$ is $\dfrac{9-6}{6-9} = -1$, so the tangent is $y - 3 = -(x-3)$, that is $y = -x + 6$.

(b) Differentiate: $3x^2 + 3y^2y' = 6y + 6xy'$. Collect y': $y'(3y^2 - 6x) = 6y - 3x^2$, so $y' = \dfrac{6y-3x^2}{3y^2-6x} = \dfrac{2y-x^2}{y^2-2x} = \dfrac{x^2-2y}{2x-y^2}$.

(c) Horizontal: numerator zero, $y = \frac{1}{2}x^2$. Substitute: $x^3 + \frac{1}{8}x^6 = 3x^3$, so $\frac{1}{8}x^6 = 2x^3$, so $x^3 = 0$ or $x^3 = 16$. With $x > 0$: $x = \sqrt[3]{16} = 2\sqrt[3]{2}$, and $y = \frac{1}{2}x^2 = 2\sqrt[3]{4}$. The denominator there is $2x - y^2 = 4\sqrt[3]{2} - 4\sqrt[3]{16} = 4\sqrt[3]{2} - 8\sqrt[3]{2} \neq 0$, so the tangent really is horizontal. The point is $(2\sqrt[3]{2}, 2\sqrt[3]{4})$. (The solution $x = 0$ gives the origin, where numerator and denominator are both zero: the curve crosses itself there.)

8. $\dfrac{dV}{dt} = 4\pi r^2\dfrac{dr}{dt}$. With $\dfrac{dV}{dt} = 100$ and $r = 10$: $100 = 400\pi\dfrac{dr}{dt}$, so $\dfrac{dr}{dt} = \dfrac{1}{4\pi}$ cm/s (about 0.080 cm/s).

9. Let x be the distance from the foot to the wall and y the height of the top. Then $x^2 + y^2 = 25$ at every moment, so $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0$. When $x = 3$, $y = 4$, and $\dfrac{dx}{dt} = 0.5$: $3\cdot 0.5 + 4\dfrac{dy}{dt} = 0$, so $\dfrac{dy}{dt} = -\dfrac{3}{8} = -0.375$. The top slides down at 0.375 m/s.

10.

(a) $f'(x) = 4x^3 - 12x$, $f''(x) = 12x^2 - 12 = 12(x-1)(x+1)$. f'' changes sign at $x = \pm 1$, so the inflection points are $(-1,-4)$ and $(1,-4)$.

(b) $g'(x) = (1-x)e^{-x}$, $g''(x) = -e^{-x} - (1-x)e^{-x} = (x-2)e^{-x}$. Sign change at $x = 2$: inflection point $(2, 2e^{-2}) = \left(2, \dfrac{2}{e^2}\right)$.

11.

(a) $f'(x) = 3x^2 + 1 > 0$ for all x, so f is strictly increasing, hence one-to-one, and has an inverse.

(b) $f(1) = 3$, so $f^{-1}(3) = 1$. $f'(1) = 4$, so $(f^{-1})'(3) = \dfrac{1}{f'(1)} = \dfrac{1}{4}$.

CE-level

12.

(a) $f'(x) = 2x\ln x + x^2 \cdot \dfrac{1}{x} = x(2\ln x + 1)$. For $x > 0$: $f'(x) = 0 \iff \ln x = -\frac{1}{2} \iff x = e^{-1/2}$. Then $f(e^{-1/2}) = e^{-1} \cdot (-\frac{1}{2}) = -\dfrac{1}{2e}$. The sign of f' is that of $2\ln x + 1$: negative before, positive after, so this is a minimum: $\left(e^{-1/2}, -\dfrac{1}{2e}\right)$, or $\left(\dfrac{1}{\sqrt{e}}, -\dfrac{1}{2e}\right)$.

(b) $f''(x) = 2\ln x + 1 + x \cdot \dfrac{2}{x} = 2\ln x + 3$. Zero at $\ln x = -\frac{3}{2}$, $x = e^{-3/2}$, with a sign change (from $-$ to $+$). $f(e^{-3/2}) = e^{-3} \cdot (-\frac{3}{2}) = -\dfrac{3}{2e^3}$. Inflection point $\left(e^{-3/2}, -\dfrac{3}{2e^3}\right)$.

(c) Slope: $f'(e^{-3/2}) = e^{-3/2}(2 \cdot (-\frac{3}{2}) + 1) = -2e^{-3/2}$. Tangent line $y = -2e^{-3/2}x + b$ through the inflection point:

$$-\frac{3}{2e^3} = -2e^{-3/2} \cdot e^{-3/2} + b = -\frac{2}{e^3} + b \quad \Longrightarrow \quad b = \frac{2}{e^3} - \frac{3}{2e^3} = \frac{1}{2e^3}.$$

So the tangent meets the y-axis at $\left(0, \dfrac{1}{2e^3}\right)$.

13.

(a) $f'(x) = \cos x + \frac{1}{2} \cdot 2\cos(2x) = \cos x + \cos 2x = \cos x + 2\cos^2 x - 1 = 2\cos^2 x + \cos x - 1 = (2\cos x - 1)(\cos x + 1)$.

(b) $f'(x) = 0$ gives $\cos x = \frac{1}{2}$ or $\cos x = -1$: $x = \frac{\pi}{3}$, $x = \pi$, $x = \frac{5\pi}{3}$. Since $\cos x + 1 \geq 0$, the sign of f' is the sign of $2\cos x - 1$ (except at π): positive on $[0, \frac{\pi}{3})$, negative on $(\frac{\pi}{3}, \frac{5\pi}{3})$, positive after. Candidates for the maximum: $f(\frac{\pi}{3}) = \frac{1}{2}\sqrt{3} + \frac{1}{2} \cdot \frac{1}{2}\sqrt{3} = \frac{3}{4}\sqrt{3}$ and the endpoint $f(2\pi) = 0$. The maximum value is $\dfrac{3\sqrt{3}}{4}$.

(c) $f'(\pi) = (2 \cdot (-1) - 1)(-1 + 1) = 0$. Near π (on both sides), $2\cos x - 1 < 0$ and $\cos x + 1 > 0$, so $f'(x) < 0$ on both sides: f is decreasing through $x = \pi$ and there is no extreme. (It is a horizontal inflection point: $f''(\pi) = -\sin\pi - 2\sin 2\pi = 0$.)

14.

(a) Point of contact at $x = 1$: $f(1) = e = e \cdot 1$ (same height) and $f'(1) = e^1 = e$, the slope of the line (same slope). So the line touches the graph at $(1, e)$.

(b) Tangent at $x = a$: $y = \dfrac{1}{a}(x - a) + \ln a$. It passes through O: $0 = -1 + \ln a$, so $a = e$. The point of contact is $(e, 1)$ and k: $y = \dfrac{1}{e}x$.

15.

(a) $f_p'(x) = 4x^3 + 2px = 2x(2x^2 + p)$. For $p \geq 0$, $2x^2 + p \geq 0$ with equality only at $x = 0$ (when $p = 0$). So f_p' has the sign of x (zero only at 0): negative for $x < 0$, positive for $x > 0$. Exactly one extreme, a minimum at $(0, 0)$.

(b) For $p < 0$: $2x^2 + p = 0$ gives $x = \pm\sqrt{-\frac{p}{2}}$. Then $x^2 = -\frac{p}{2}$ and $f_p(x) = x^4 + px^2 = \frac{p^2}{4} - \frac{p^2}{2} = -\frac{p^2}{4}$. The minima are $\left(\pm\sqrt{-\frac{p}{2}}, -\frac{p^2}{4}\right)$. (The third extreme is a maximum at $(0, 0)$.)

(c) At the minima $p = -2x^2$. Then $y = x^4 + px^2 = x^4 - 2x^4 = -x^4$.

16.

(a) $f_p'(x) = e^{-px} - pxe^{-px} = (1-px)e^{-px}$. Zero at $x = \frac{1}{p}$ with a sign change (the factor $1-px$ changes sign, $e^{-px} > 0$). $f_p(\frac{1}{p}) = \frac{1}{p}e^{-1} = \dfrac{1}{pe}$. Extreme $\left(\dfrac{1}{p}, \dfrac{1}{pe}\right)$ (a maximum for $p > 0$, a minimum for $p < 0$).

(b) At the extreme $y = \dfrac{1}{e} \cdot \dfrac{1}{p} = \dfrac{1}{e}x$. All extremes lie on the line $y = \dfrac{x}{e}$.

(c) $f_p''(x) = -pe^{-px} - p(1-px)e^{-px} = p(px-2)e^{-px}$. It changes sign at $x = \frac{2}{p}$. So $\frac{2}{p} = 4$, $p = \frac{1}{2}$.

17.

(a) Volume: $\pi r^2 h = 330$, so $h = \dfrac{330}{\pi r^2}$. Surface: $S = 2\pi r^2 + 2\pi rh = 2\pi r^2 + 2\pi r \cdot \dfrac{330}{\pi r^2} = 2\pi r^2 + \dfrac{660}{r}$.

(b) $S'(r) = 4\pi r - \dfrac{660}{r^2} = 0 \iff r^3 = \dfrac{165}{\pi} \iff r = \sqrt[3]{165/\pi} \approx 3.74$ cm. S' changes from $-$ to $+$ there, so it is a minimum. Then $h = \dfrac{330}{\pi r^2} = \dfrac{2\pi r^3}{\pi r^2} = 2r$ (using $330 = 2\pi r^3$ from $r^3 = 165/\pi$). So the height equals the diameter, about 7.49 cm. (Real cans are taller because the top and bottom are made of thicker metal.)

18. $P = (x, x^2)$. Minimise $d^2 = x^2 + (x^2-3)^2$. Derivative: $2x + 2(x^2-3) \cdot 2x = 2x(2x^2-5)$. Zeros: $x = 0$ ($d^2 = 9$) and $x = \pm\sqrt{\frac{5}{2}}$, where $d^2 = \frac{5}{2} + \frac{1}{4} = \frac{11}{4}$. The sign chart of $2x(2x^2-5)$ is $-, +, -, +$, so $x = \pm\sqrt{5/2}$ are minima and $x = 0$ is a local maximum. The smallest distance is $\sqrt{\frac{11}{4}} = \frac{1}{2}\sqrt{11}$, at $P = (\pm\frac{1}{2}\sqrt{10}, \frac{5}{2})$.

19.

(a) By similar triangles the water surface has radius $r = \frac{2}{4}h = \frac{1}{2}h$. So $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \cdot \frac{1}{4}h^2 \cdot h = \dfrac{\pi h^3}{12}$.

(b) $\dfrac{dV}{dt} = \dfrac{\pi h^2}{4}\dfrac{dh}{dt}$. With $\dfrac{dV}{dt} = 0.5$ and $h = 1$: $\dfrac{dh}{dt} = \dfrac{0.5 \cdot 4}{\pi} = \dfrac{2}{\pi}$ m/min (about 0.64 m/min).

20.

(a) Long division: $f(x) = x + 1 + \dfrac{4}{x-1}$. Vertical asymptote $x = 1$ (the denominator is zero and the numerator is $4 \neq 0$). As $x \to \pm\infty$, $\dfrac{4}{x-1} \to 0$, so the oblique asymptote is $y = x + 1$. No horizontal asymptote.

(b) $f'(x) = 1 - \dfrac{4}{(x-1)^2} = \dfrac{(x-1)^2 - 4}{(x-1)^2} = \dfrac{(x-3)(x+1)}{(x-1)^2}$. Sign chart: $+$ for $x < -1$, $-$ on $(-1, 1)$ and $(1, 3)$, $+$ for $x > 3$. Maximum $f(-1) = \dfrac{4}{-2} = -2$; minimum $f(3) = \dfrac{12}{2} = 6$.

(c) $f''(x) = \dfrac{8}{(x-1)^3}$, which is never zero. (It changes sign at $x = 1$, but 1 is not in the domain, so that is not an inflection point.)

(d) On $(-\infty, 1)$, f rises to -2 and then falls to $-\infty$; on $(1, \infty)$, f falls from $+\infty$ to 6 and then rises to $+\infty$. Range: $(-\infty, -2] \cup [6, \infty)$.

21.

(a) $2x - (y + xy') + 2yy' = 0$, so $y'(2y - x) = y - 2x$ and $y' = \dfrac{y - 2x}{2y - x}$.

(b) $(2, 1)$ is on K: $4 - 2 + 1 = 3$. There the denominator is $2 - 2 = 0$ and the numerator $1 - 4 = -3 \neq 0$, so the slope is infinite: the tangent line is vertical (it is $x = 2$). A cleaner way to say this: treat x as a function of y; then $\dfrac{dx}{dy} = \dfrac{2y - x}{y - 2x} = 0$ at $(2, 1)$.

(c) At the highest point the tangent is horizontal: $y - 2x = 0$, $y = 2x$. Substitute: $x^2 - 2x^2 + 4x^2 = 3x^2 = 3$, so $x = \pm 1$, giving $(1, 2)$ and $(-1, -2)$. The denominator $2y - x$ is 3 and -3 there, not zero. K is a closed bounded curve (an ellipse), so it has a highest point, and it must be one of these: $(1, 2)$.

22.

(a) $f'(x) = 1 - 2\sin x = 0 \iff \sin x = \frac{1}{2} \iff x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$. f' goes from $+$ to $-$ at $\frac{\pi}{6}$ (maximum) and from $-$ to $+$ at $\frac{5\pi}{6}$ (minimum). Maximum $\left(\frac{\pi}{6}, \frac{\pi}{6} + \sqrt{3}\right)$, minimum $\left(\frac{5\pi}{6}, \frac{5\pi}{6} - \sqrt{3}\right)$.

(b) $f''(x) = -2\cos x$, which changes sign at $x = \frac{\pi}{2}$, so there is an inflection point at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ (since $\cos\frac{\pi}{2} = 0$). Slope there: $f'(\frac{\pi}{2}) = 1 - 2 = -1$. Tangent: $y = -(x - \frac{\pi}{2}) + \frac{\pi}{2} = -x + \pi$. At $x = \pi$: $y = 0$. So the line passes through $(\pi, 0)$.

Challenge

23. Differentiate $y^q = x^p$ with respect to x: $qy^{q-1}y' = px^{p-1}$. So

$$y' = \frac{p}{q} \cdot \frac{x^{p-1}}{y^{q-1}} = \frac{p}{q} \cdot \frac{x^{p-1}}{x^{(p/q)(q-1)}} = \frac{p}{q} x^{p-1-p+p/q} = \frac{p}{q} x^{p/q-1}.$$

(Note $y > 0$ for $x > 0$, so we never divide by zero. This argument assumes y is differentiable. That is justified: $y = \sqrt[q]{x^p}$, and the q-th root is the inverse of $t \mapsto t^q$, whose derivative qt^{q-1} is not zero for $t > 0$; by notes section 7 the root is differentiable, and then so is y by the chain rule.)

24.

(a) tan is increasing on $(-\frac{\pi}{2}, \frac{\pi}{2})$ with inverse arctan. If $y = \arctan x$ then $\tan y = x$, and by the inverse-function rule $(\arctan x)' = \dfrac{1}{(\tan)'(y)} = \dfrac{1}{1 + \tan^2 y} = \dfrac{1}{1 + x^2}$.

(b) $h'(x) = \dfrac{1}{1 + x^2} + \dfrac{1}{1 + x^{-2}} \cdot \left(-\dfrac{1}{x^2}\right) = \dfrac{1}{1 + x^2} - \dfrac{1}{x^2 + 1} = 0$. So h is constant on $(0, \infty)$: $h(x) = h(1) = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$. (Geometrically: in a right triangle with legs 1 and x, the two acute angles add up to $\frac{\pi}{2}$.)

(c) The derivative is still 0 on $(-\infty, 0)$, but that is a separate interval, so the constant may differ. $h(-1) = -\frac{\pi}{4} - \frac{\pi}{4} = -\frac{\pi}{2}$. So $h(x) = -\frac{\pi}{2}$ for $x < 0$. Lesson: "derivative zero implies constant" only holds on an interval.

25.

(a) $x^x = e^{x \ln x}$, so $f'(x) = e^{x \ln x}(\ln x + 1) = x^x(\ln x + 1)$.

(b) $f'(x) = 0 \iff \ln x = -1 \iff x = \frac{1}{e}$, with sign change $-$ to $+$. Minimum value $f(\frac{1}{e}) = \left(\frac{1}{e}\right)^{1/e} = e^{-1/e} \approx 0.692$.

(c) $x \ln x \to 0$ as $x \to 0^+$ (the logarithm goes to $-\infty$ much more slowly than x goes to 0; one way to see it: put $x = e^{-t}$, then $x \ln x = -te^{-t} \to 0$ as $t \to \infty$). So $f(x) = e^{x \ln x} \to e^0 = 1$.

26. Touching at $x = c$ requires $a^c = c$ and $a^c \ln a = 1$. Substituting the first in the second: $c \ln a = 1$. But $a^c = e^{c \ln a} = e^1 = e$, so $c = e$ and $\ln a = \frac{1}{e}$, so $a = e^{1/e} \approx 1.4447$. Point of contact (e, e).

Counting: let $g(x) = a^x - x$. Then $g'(x) = a^x \ln a - 1$ is increasing, so g is convex with one minimum, at x_0 where $a^{x_0} \ln a = 1$. $g \to \infty$ at both ends (as $x \to -\infty$ because of $-x$, as $x \to \infty$ because the exponential beats x). So the number of zeros is 2, 1 or 0 as the minimum value is negative, zero or positive.

- If $1 < a < e^{1/e}$: $g(e) = a^e - e < \left(e^{1/e}\right)^e - e = 0$, so the minimum is negative: two solutions.
- If $a = e^{1/e}$: the touching case, one solution $x = e$.
- If $a > e^{1/e}$: $a^x > e^{x/e}$ for $x > 0$, and $e^{x/e} \geq x$ for all x (the graph of $e^{x/e}$ lies above its tangent line $y = x$ at $x = e$, because it is convex). So $a^x > x$ for $x > 0$, and for $x \leq 0$ trivially $a^x > 0 \geq x$: no solutions.

Conclusion: two solutions for $1 < a < e^{1/e}$, one solution for $a = e^{1/e}$, none for $a > e^{1/e}$. (A numerical check: $a = 1.2$ gives two solutions, $a = 1.5$ none.)

27.

(a) $\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0$, so $y' = -\dfrac{x^{-1/3}}{y^{-1/3}} = -\left(\dfrac{y}{x}\right)^{1/3}$. At (x_0, y_0) this is $-\left(\dfrac{y_0}{x_0}\right)^{1/3}$.

(b) Tangent: $y - y_0 = m(x - x_0)$ with $m = -y_0^{1/3}x_0^{-1/3}$. Setting $y = 0$: $x = x_0 - \dfrac{y_0}{m} = x_0 + y_0^{2/3}x_0^{1/3} = x_0^{1/3}\left(x_0^{2/3} + y_0^{2/3}\right) = x_0^{1/3}$. Setting $x = 0$: $y = y_0 - mx_0 = y_0 + y_0^{1/3}x_0^{2/3} = y_0^{1/3}\left(y_0^{2/3} + x_0^{2/3}\right) = y_0^{1/3}$.

(c) The length is $\sqrt{x_0^{2/3} + y_0^{2/3}} = \sqrt{1} = 1$.

28. $x = 0$ is never a solution ($e^0 = 1 \neq 0$). Look at the line $y = px$ against the graph of $y = e^x$.

- $p < 0$: $g(x) = e^x - px$ has $g'(x) = e^x - p > 0$, so g is increasing; $g \to -\infty$ as $x \to -\infty$ and $g \to \infty$ as $x \to \infty$. Exactly **one** solution (a negative one).
- $p = 0$: $e^x = 0$ has **no** solution.
- $p > 0$: solutions have $x > 0$, and $e^x = px \iff p = \dfrac{e^x}{x}$. The function $k(x) = \dfrac{e^x}{x}$ on $(0, \infty)$ has $k'(x) = \dfrac{(x-1)e^x}{x^2}$, minimum $k(1) = e$, and $k \to \infty$ at both ends. So: **no** solution for $0 < p < e$, **one** for $p = e$ (the touching line from notes example 10.2), **two** for $p > e$.

29.

(a) $\dfrac{f(h) - f(0)}{h} = h \sin \dfrac{1}{h}$. Since $\left|\sin \frac{1}{h}\right| \leq 1$, this lies between $-|h|$ and $|h|$, so it tends to 0. Thus $f'(0) = 0$.

(b) For $x \neq 0$: $f'(x) = 2x \sin \dfrac{1}{x} + x^2 \cos \dfrac{1}{x} \cdot \left(-\dfrac{1}{x^2}\right) = 2x \sin \dfrac{1}{x} - \cos \dfrac{1}{x}$.

(c) The first term tends to 0. The second, $\cos \frac{1}{x}$, keeps taking the values 1 (at $x = \frac{1}{2k\pi}$) and -1 (at $x = \frac{1}{(2k+1)\pi}$) arbitrarily close to 0. So $f'(x)$ takes values near -1 and near 1 arbitrarily close to 0 and has no limit. f' exists everywhere but is not continuous at 0.

(d) $f(x) = 0$ at every $x = \frac{1}{k\pi}$, and these points pile up at 0. If f is the inner function and we look at $x = 0$, then $\Delta u = f(h) - f(0) = 0$ for $h = \frac{1}{k\pi}$, however small h is. The naive proof divides by Δu and fails for these h. The proof with the helper function Q in notes section 3 handles it without trouble.